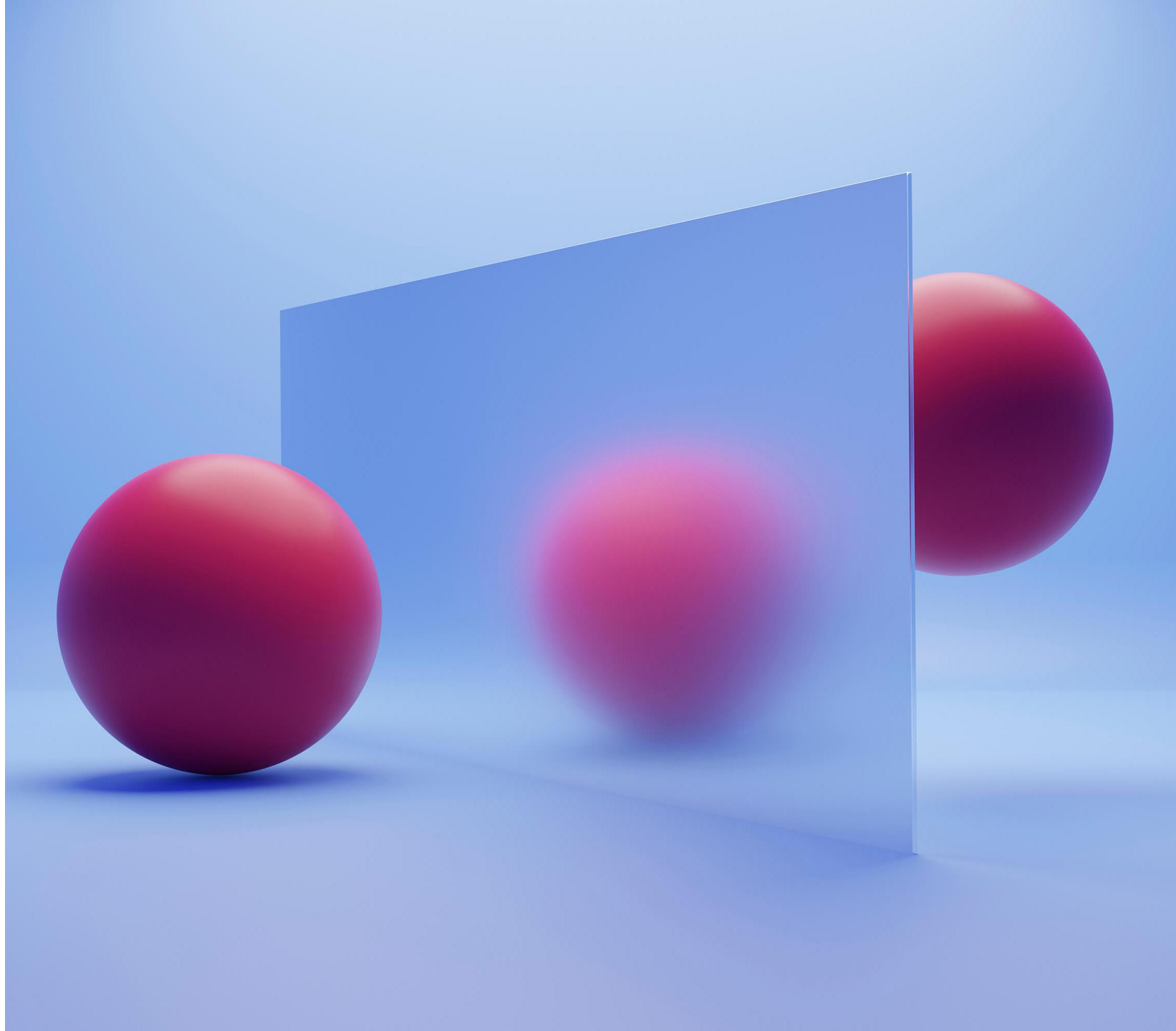


# Project Portfolio

Muhammad Abdullah  
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abdullah8t90@gmail.com



# Education

- **MS Structural Engineering (2024-2026)**  
University of Illinois Urbana-Champaign

Structural Dynamics, Mechanics, FEM, Nonlinear FEM, Concrete Design, Advanced Steel Design, Structural Damping, Reliability Analysis

- **BE Civil Engineering (2017-2021)**  
MCE, NUST

Steel-1, Design of Concrete Structures, Structural Analysis, Pavement Engineering, Footing Design

# Technical Skills

- ETABS, SAP2000, Excel, Revit, AutoCAD, Primavera, QGIS, HEC-RAS
- Python, MATLAB, Grasshopper

# Interests

- Optimization, Automation, Computational design

# Experience

- **Graduate Researcher (Mar 2025 – May 2026)**  
Smart Structures Technology Lab, UIUC

Integrated reliability constraints into a legacy stochastic topology optimization framework

Implemented and validated adjoint sensitivity formulations to compute gradients for highly non-linear reliability constraints

Refactored core MATLAB scripts into a modular architecture to allow for greater flexibility in adding components to the framework

- **Management Trainee Engineer**  
NLC (Feb 2022 – Mar 2023)

Managed execution of structural and road components in highway projects and inspection of bridge elements

Responsible for supervising sub-contractors, and coordinating between consultant and contractor staff

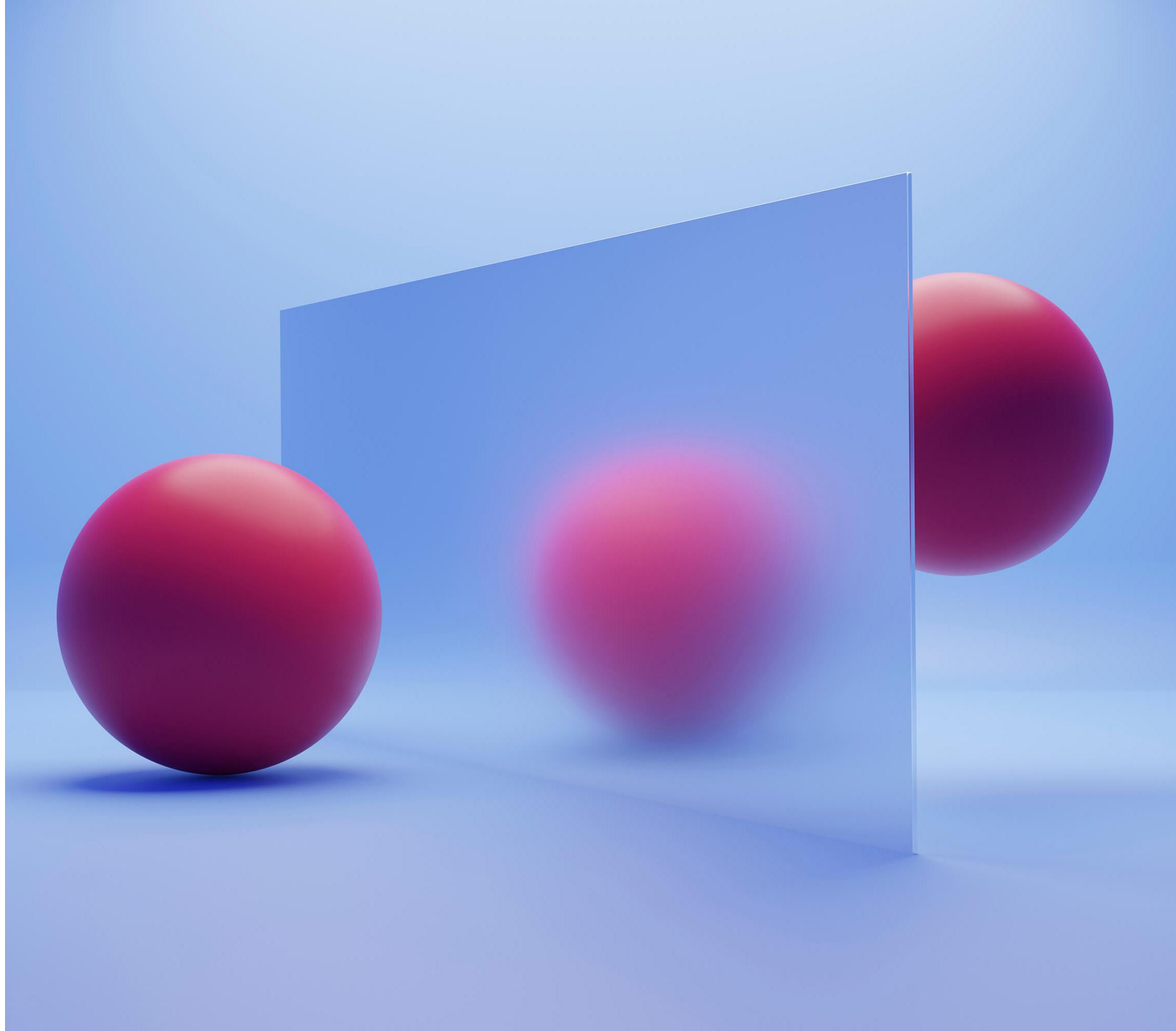
Prepared daily work reports, presentation materials for project director briefings, and updated monthly work schedules and progress with baselines in Primavera

Prepared bar bending schedules for building construction projects, and coordinated with main office for procurement and delivery of materials

- **Intern (Nov 2021 –Jan 2022)**  
Designment Consulting

Assisted with inspection of site-work

# **Academic Projects**



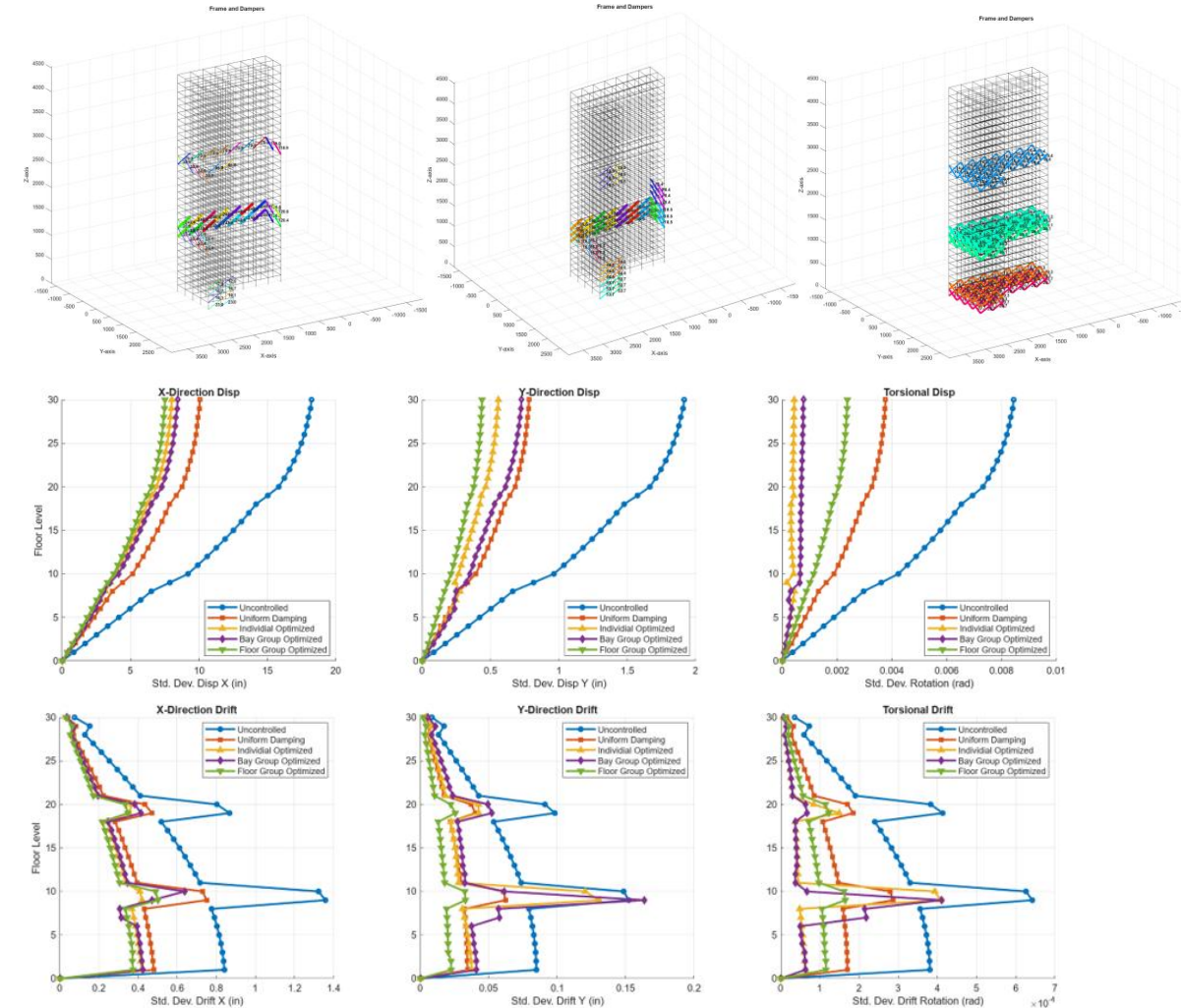
# Optimizing Viscous Damper Placement for Irregular L-Shaped Tall Buildings

(Term Project – Structural Damping)

Developed a framework for optimizing rotational performance of asymmetric buildings under stochastic excitation, with damping coefficients as design variables.

Tested individual/grouped/floor-grouped damper configurations to analyze tradeoffs in performance vs practicality

Created a pipeline to extract and process structural matrices from models developed in SAP2000 to MATLAB, applying rigid body constraints to floors and performing reduction to a simpler structure, and then using a gradient based optimizer.



# Life-Cycle Cost Assessment in a Reliability Framework

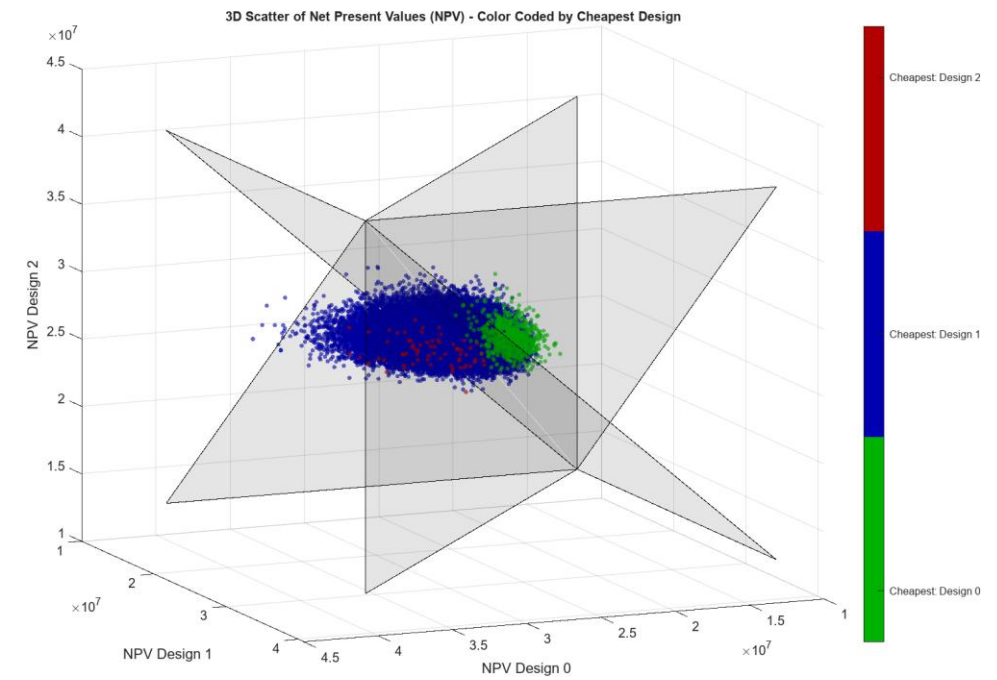
(Term Project - Reliability Analysis)

Used structural reliability analysis methods (FORM/SORM) for decision making between various design alternatives for a pavement project based on life cycle costs, by treating costs as random variables

Validated results using Monte Carlo analysis

Plotted sensitivities of choice to cost variables and parameters

Used MATLAB for analysis and validation pipelines

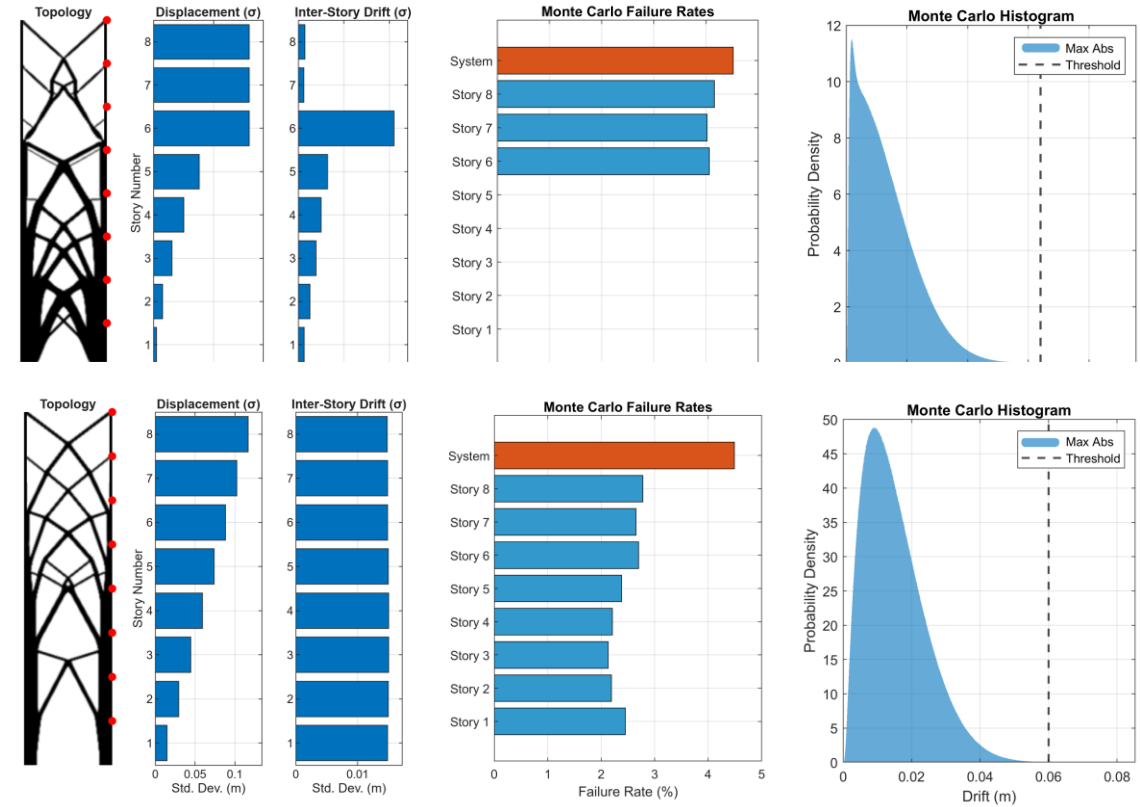


# Reliability Constraints in a Stochastic Topology Optimization Framework (Thesis)

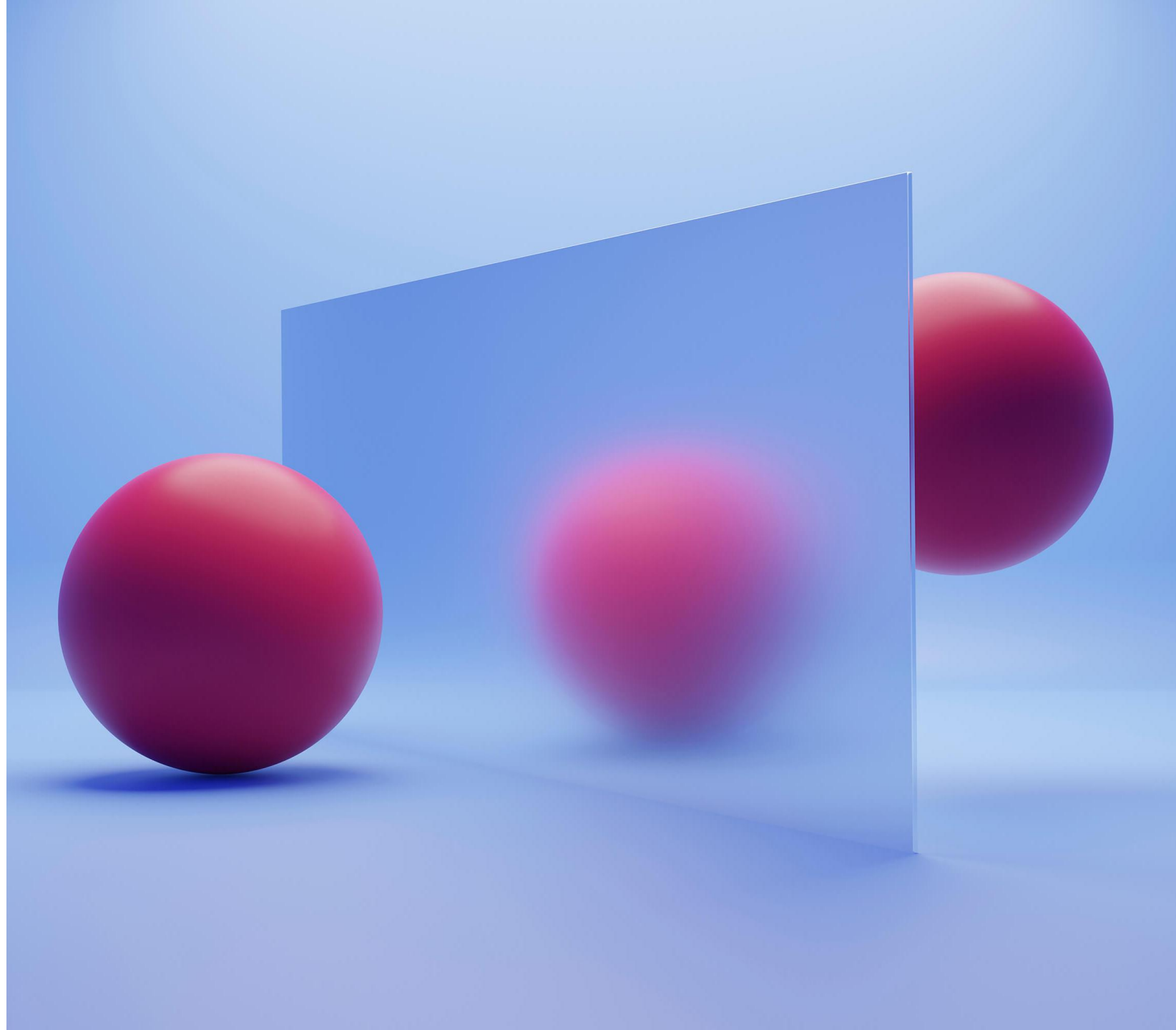
Implemented reliability constraints (component outcrossing rate and first passage failure rate) and evaluated gradient expressions for use in a stochastic topology optimization framework

Developed and implemented expressions for system outcrossing rate as constraints for global control of optimization

Resulted in topologies with very low probability of exceeding specified drift or displacement thresholds under stochastic base excitations. Constraints were set at 5% probability of exceedance under 60 seconds of ground motion



# **Professional Experience**



# Sheikh Badin Highway Project (Package I - II)

Managed execution of bridge components (girders, piles) and culverts

Oversaw excavation/fill and pavement construction

Responsible for supervising sub-contractors, and coordinating between consultant and contractor staff

Prepared daily work reports and presentation materials for briefings.



# Proposal for Smaller Culvert

Original design received for culvert at RD 11+500 was a 4x4m 3-cell culvert

Unable to properly build and pour due to access issues, also deemed to be excessively over specified

Alternative proposal was sent arguing that the catchment for the channel did not warrant such a large culvert and by providing a simple steady state simulation in HEC-RAS of channel flow to validate the argument

Proposal was accepted and an updated design issued for a single cell 4x4m culvert, which significantly saved time and cost



# Lakki Marwat Nursing Campus and Paramedics Hospital

Prepared bar bending schedules for buildings,  
and coordinated with main office for  
procurement and delivery of materials

Responsible for supervising sub-contractors and  
inspecting rebar layout on site, and coordinating  
between consultant and contractor staff

Prepared daily work reports and presentation  
materials for briefings

Updated monthly work schedules and progress  
with baselines in Primavera

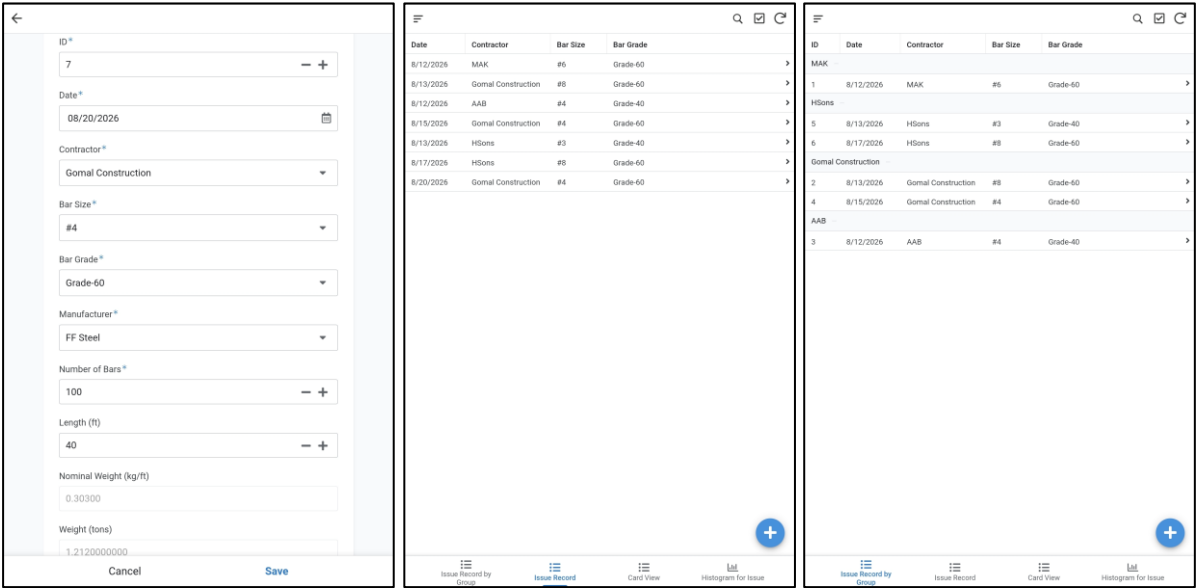


# Steel Inventory App

Used AppSheet to create an android app linked to Google Sheets for inventory management as an alternative to physical record keeping

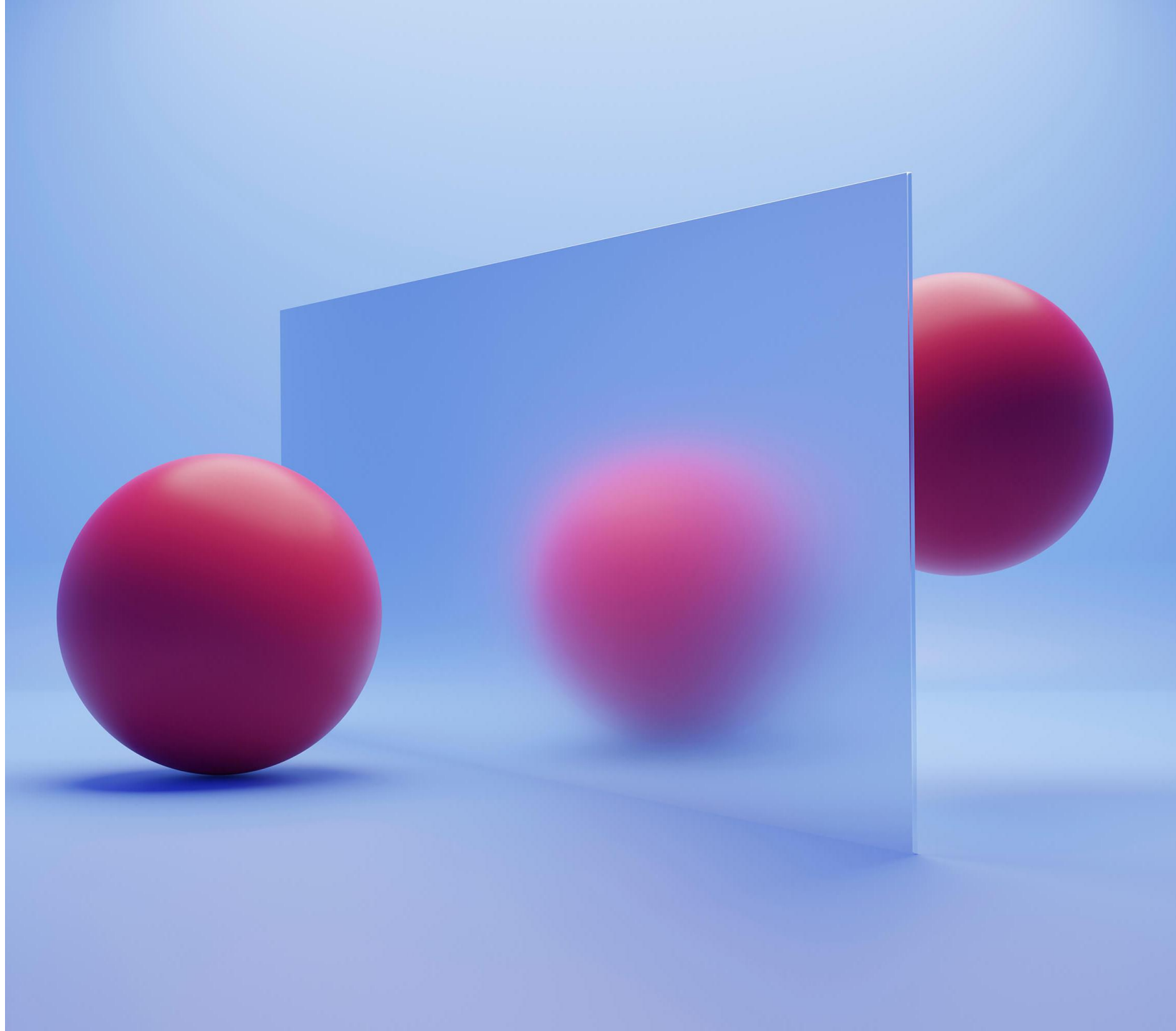
Rebar issued to subcontractors was recorded with relevant data fields

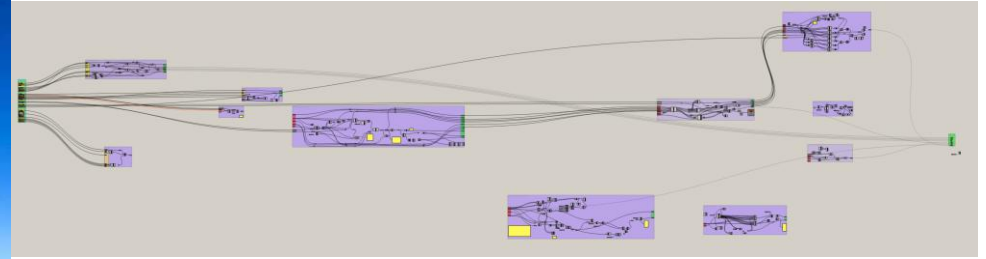
Allowed for easier data entry and subsequent processing for weekly and monthly reports and inventory management and tracking



Demo Data

# **Additional Personal Projects**

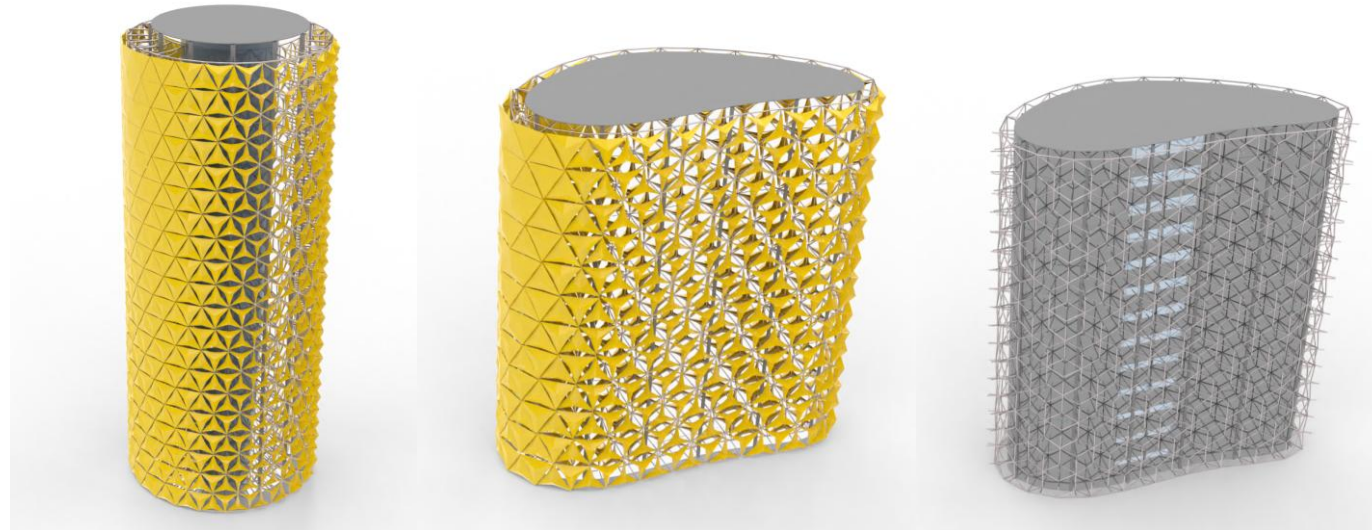




# Grasshopper

Inspired by the Mashrabiya facade of the Al-Bahar towers

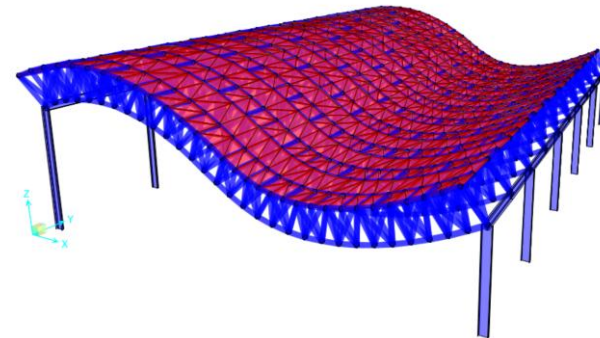
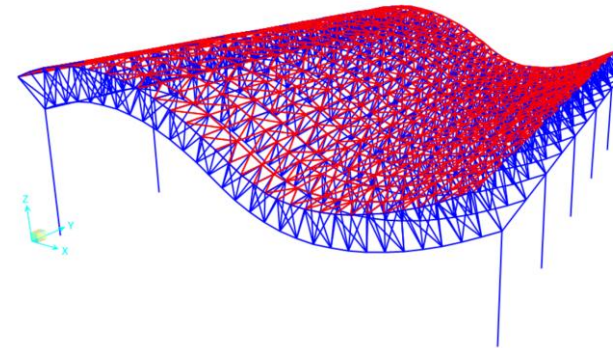
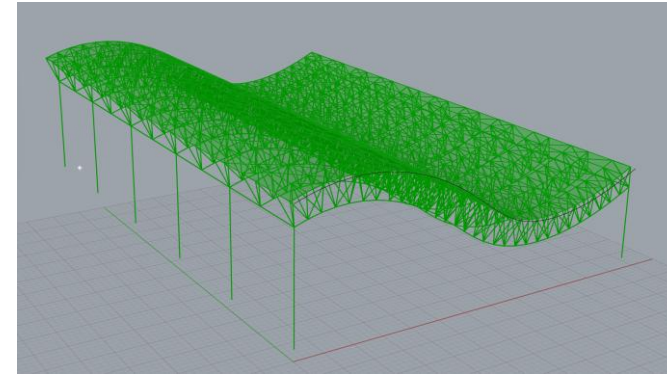
Created an algorithm in grasshopper to recreate the façade parametrically



# Grasshopper to SAP2000

Used BHoM (Building and Habitats object Model) to explore opportunities for porting parametric designs between SAP2000 and grasshopper

Image shows a truss supported roof that takes a profile curve. Elements are structured in grasshopper and exported to SAP2000 for further analysis and design.



# Thank you

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